

Applied Linear Algebra (EE 635): Problem Set 5

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Questions

1. The set of all continuous real-valued functions defined on a closed interval $[a, b]$ in $\mathbb{R}$ is denoted by $C[a, b]$. This set is a subspace of the vector space of all real-valued functions defined on $[a, b]$.
 - (a) What facts about continuous functions should be proved in order to demonstrate that $C[a, b]$ is indeed a subspace as claimed?
 - (b) Show that $\{f \in C[a, b] \mid f(a) = f(b)\}$ is a subspace of $C[a, b]$.

2. Let

$$H = \left\{ \begin{bmatrix} s \\ 3s \\ 2s \end{bmatrix} : s \in \mathbb{R} \right\}.$$

Find a vector $v \in \mathbb{R}^3$ such that

$$H = \text{Span}\{v\}.$$

Why does this show that H is a subspace of $\mathbb{R}^3$?

3. Let W be the set of all vectors of the form shown below, where a, b , and c represent arbitrary real numbers. In each case, either find a set S of vectors that spans W or give an example to show that W is not a vector space.

- (a)

$$\begin{bmatrix} 3a + b \\ 4 \\ a - 5b \end{bmatrix}$$

- (b)

$$\begin{bmatrix} -a + 1 \\ a - 6b \\ 2b + a \end{bmatrix}$$

(c)

$$\begin{bmatrix} a-b \\ b-c \\ c-a \\ b \end{bmatrix}$$

(d)

$$\begin{bmatrix} 4a+3b \\ 0 \\ a+b+c \\ c-2a \end{bmatrix}$$

4. Let P_n denote the vector space of all polynomials with real coefficients of degree at most n . For each of the following sets, determine if it is a subspace of P_n for an appropriate value of n . Justify your answers.
- (a) All polynomials of the form $p(t) = at^2$, where $a \in \mathbb{R}$.
 - (b) All polynomials of the form $p(t) = a + t^2$, where $a \in \mathbb{R}$.
 - (c) All polynomials of degree at most 3 with integer coefficients.
 - (d) All polynomials $p \in P_n$ such that $p(0) = 0$.
5. Let V and W be finite-dimensional vector spaces over $\mathbb{R}$, with $\dim V = n$ and $\dim W = m$. Consider the set

$$\mathcal{L}(V, W) = \{T \mid T : V \rightarrow W \text{ is linear}\}.$$

- (a) Show that $\mathcal{L}(V, W)$ is a vector space over $\mathbb{R}$.
 - (b) Construct a basis for $\mathcal{L}(V, W)$ and determine its dimension.
6. Consider the following problems related to linear independence.
- (a) Let $p_1(t) = 1 + t^2$ and $p_2(t) = 1 - t^2$. Determine whether $\{p_1, p_2\}$ forms a linearly independent set in P_3 . Give reasons for your answer.
 - (b) Show that the set of functions

$$\{t, \sin t, \cos 2t, \sin t \cos t\}$$

is linearly independent on $\mathbb{R}$. Start by assuming that

$$c_1 t + c_2 \sin t + c_3 \cos 2t + c_4 \sin t \cos t = 0 \quad \text{for all } t \in \mathbb{R}, \quad (1)$$

and show that $c_1 = c_2 = c_3 = c_4 = 0$. (Hint: evaluate (1) at carefully chosen values of t , for example $t = 0, \frac{\pi}{2}, \pi, \frac{\pi}{4}$, to obtain a linear system for the coefficients.)